doc-agent: codebase guide

An Agentic-RAG system over scanned documents. Scanned page-images go in; a grounded, cited answer (or an honest abstention) comes out. This guide explains what each module does and how the pieces connect.

1. The big picture

```
BUILD (pipeline.build_knowledge_base)            QUERY (pipeline.answer)
 data/raw/*.png                                   question
   |  loader.load_pages    (Stage 1)               |  Retriever (dense, faiss)   (Stage 5)
   |  preprocess.run       deskew/denoise/CLAHE    v
   |  enhance.run          denoiser (Stage 1)     Agent.run loop (Stage 6):
   |  layout.detect        text-line boxes (2)      ON_STEP -> decide() -> ON_TOOL_CALL
   |  ocr.transcribe       TrOCR / sidecar (3)      -> act() (tool) -> observe ... repeat
   |  AFTER_OCR hook       PII redaction            BEFORE_ANSWER hook (grounding, PII)
   |  chunk.split          sliding window (4)       synthesize() -> LLM -> Answer
   |  embed.encode         MiniLM vectors           AFTER_ANSWER hook (trace)
   |  store.build          faiss index on disk
   v
 data/index/index.faiss + chunks.jsonl  ---------->  loaded by retrieval
```

The stage order lives in `pipeline.py` and must not change. Horizontal concerns (tracing, PII, security, grounding) are **not** inlined into stages; they attach at fixed hook seams (section 3).

2. Module-by-module reference

Module	Status	Role
`contracts.py`	FIXED	Pydantic data contracts passed between stages: Page, Region, Chunk (has a retrieval score), Query, Citation, Answer, ToolResult, TraceStep.
`config.py / settings.py`	FIXED	config.load() reads configs/config.yaml (all models and parameters). settings reads secrets (LLM_API_KEY, WANDB_API_KEY) from .env.
`pipeline.py`	FIXED	build_knowledge_base(cfg) runs Stages 1-4 and writes the index; answer(query, cfg) builds a Retriever and runs the Agent.
`hooks.py`	FIXED	The seam registry. register(seam, handler) and run(seam, ctx). Nine seams: after_ingest, after_ocr, before_index, after_retrieve, on_step, on_tool_call, before_answer, after_answer, on_log.
`wiring.py`	FIXED	register_all(cfg): clears hooks and wires tracing, PII, guardrails and grounding. Called at the start of every build and every answer.
`logging_conf.py`	IMPL	get_logger (JSON logs). register() adds the audit trail: every decide / tool call / answer is appended as a TraceStep line to traces/run.jsonl (PII-scrubbed via ON_LOG).
`ingest/loader.py`	IMPL	load_pages: scans data/raw for images, builds Page objects (doc_id from folder or filename), validates and records the corpus version. data_paths(cfg) resolves every directory.
`ingest/preprocess.py`	IMPL	Classical cleanup: skew estimate + rotation, median denoise, CLAHE contrast. Also writes per-page quality stats (blur, contrast, bucket) to data/interim/page_meta.json for fairness/drift.
`ingest/enhance.py`	IMPL	DenoiseNet (small residual encoder-decoder) + Enhancer.train/apply. Uses a trained checkpoint if models/enhancer.pt exists, else classical non-local-means.
`vision/layout.py`	IMPL	Classical OpenCV layout detection: Otsu ink mask, horizontal dilation, contours -> text-line Regions (kind = text / heading / figure), in reading order. Fills PAGE_META for OCR.
`vision/ocr.py`	IMPL	Reader + transcribe: Regions -> line-level Chunks. backend 'auto' uses a <page_id>.txt sidecar when present, otherwise TrOCR on the cropped region.
`index/chunk.py`	IMPL	split: merges line chunks per document and cuts sliding windows of chunk_tokens words with overlap; records every page a chunk touches.
`index/embed.py`	IMPL	Cached SentenceTransformer (MiniLM); encode/encode_texts return L2-normalised float32 vectors (inner product = cosine).
`index/store.py`	IMPL	build/load a faiss index (flat or HNSW) plus chunks.jsonl in data/index/.

Module	Status	Role
`retrieval/retriever.py`	IMPL	Retriever.retrieve embeds the query, searches faiss, sets chunk.score = cosine. top_score / is_weak / next_k (provided) define the evidence-strength policy.
`retrieval/rerank.py`	IMPL	Optional cross-encoder reorder (cfg retrieve.rerank). Degrades gracefully if the model cannot load. Never changes chunk.score.
`llm/client.py`	IMPL	The only place LLMs are called. Anthropic or OpenAI by key; with no key, an offline extractive fallback that quotes the best-matching evidence sentence.
`llm/prompts.py`	IMPL	All prompt templates: DECIDE (query rewrite), SYNTHESIZE (grounded + citation + abstain token), JUDGE.
`llm/postprocess.py`	IMPL	format_answer (citations, grounded flag, abstention), support_score, and the BEFORE_ANSWER grounding gate that marks state['abstain'] when evidence is weak.
`agent/agent.py`	IMPL (loop FIXED)	The agent. run() loop is fixed; decide() implements evidence-gated re-search (retrieve -> if weak widen k and reword query -> else answer; abstain at k_max); act() dispatches tools; synthesize() builds the cited answer.
`agent/tools.py`	IMPL	Nine locked tools: retrieve, rerank, read_page, enhance_page, extract, aggregate, cite, calculator (safe AST), escalate_to_human. bind() gives them the retriever.
`agent/guardrails.py`	IMPL	Guardrails.check at ON_TOOL_CALL: unknown tool, step and cost budgets, autonomy level for side-effect tools, prompt-injection in arguments. sanitize() masks injection phrases in document text.
`agent/memory.py`	IMPL	Working memory: add() observations, recall() by word overlap.
`agent/hitl.py, hitl_store.py`	IMPL	Human-in-the-loop: escalate() enqueues a review item (SQLite) and returns 'pending' (action blocked); pending(), resolve(), decision().
`governance/pii.py`	IMPL	detect/redact emails, phones, SSN, cards, long IDs. register() scrubs chunks (AFTER_OCR), agent evidence (BEFORE_ANSWER) and log records (ON_LOG).
`data/validate.py, versioning.py`	IMPL	validate (structure), split_by_doc (no document leakage), check_floors (page/word minimums); snapshot hashes the corpus into data/versions.jsonl.
`serve/api.py, ui.py`	IMPL	FastAPI POST /answer (+latency log) and GET /health; Gradio UI showing answer, status and cited sentences.
`eval/metrics.py`	IMPL	ocr_f1, recall_at_k, groundedness, citation_accuracy, ece, subgroup_gap.
`eval/calibration.py`	IMPL	ECE and temperature scaling (NLL-fitted).
`eval/judge.py`	IMPL	LLM-as-judge for open answers; transparent heuristic when offline.
`eval/ablation.py`	IMPL	load_tasks, evaluate (full-agent metrics) and run (ablations: no rerank, no re-search, small k).
`eval/fairness.py, robustness.py, interpret.py`	IMPL	Per-scan-quality subgroup audit; corruption stress test; why-retrieved / where-looked explanation (no Grad-CAM).
`mlops/tracking.py, registry.py, monitor.py`	IMPL	W&B (disabled without key) + local metrics log; checkpoint registry with sha256; drift + latency-SLO monitor.
`rl/env.py, policy.py, rlvr.py, train_rl.py`	IMPL	Simulated tool-selection Gym env (answer / re-search / abstain), actor-critic policy, exact-match verifiable reward, PPO training.
`training/*`	IMPL	Lightning DataModule (document-level split), LitComponent (enhancer), train.main, LoRA (peft) and dynamic quantization helpers.
`optional/*`	IMPL	Off by default: API auth + rate limit, cache, config snapshot, DP-style optimizer, query expansion, retrain trigger, streaming ingest, synthetic text-lines.

3. Cross-cutting features and hook seams

Feature	Owner	Seam(s)	Behaviour
Tracing / audit	logging_conf.register	ON_STEP, ON_TOOL_CALL, AFTER_ANSWER	Appends TraceStep JSON to traces/run.jsonl: the path depends on observed top_score.
PII / privacy	governance/pii.register	AFTER_OCR, BEFORE_ANSWER, ON_LOG	Redacts before indexing, in evidence and in logs. Side effect: PII cannot be answered from the corpus.
Security	agent/guardrails.register	ON_STEP, ON_TOOL_CALL	Budgets, autonomy, injection detection; budgets reset each query.
Grounding	llm/postprocess.register	BEFORE_ANSWER	No strong evidence -> abstain instead of fabricate.

Handlers mutate the context in place (chunks, agent state, answer, log record), because the pipeline and agent ignore hooks.run()'s return value.

4. The agent loop in detail

```
state = {query, obs: []}
repeat up to agent.max_steps:
    ON_STEP hook                      (trace; guardrail budget reset on the first step)
    action = decide(state)
      - no observations      -> retrieve(query, k)
      - top_score >= weak_threshold -> stop (answer)
      - else k2 = next_k(k):  k2 -> retrieve(reworded query, k2)   (widen + reformulate)
                              None -> state.abstain = True, stop
    ON_TOOL_CALL hook                 (guardrails check + trace)
    result = act(action)              (runs the tool; failures become ToolResult(ok=False))
BEFORE_ANSWER hook                    (PII scrub of evidence; grounding gate)
synthesize(state):  abstain? -> "Insufficient evidence"
                    else prompt LLM with top n_context chunks (sanitised),
                    keep only valid [chunk_id] citations, check lexical support,
                    confidence = 0.5*top_score + 0.5*support
AFTER_ANSWER hook                     (trace)
```

5. Configuration (configs/config.yaml)

Key	Meaning
enhance / layout / ocr	Stage models and backends (ocr.backend: auto | sidecar | trocr).
embed, index	Embedding model; index type (faiss:flat | faiss:hnsw), chunk_tokens, overlap.
retrieve	k, k_step, k_max, weak_threshold (drive re-search), rerank + reranker model.
agent	max_steps, autonomy, budget_usd, cost_per_call_usd, llm_model, n_context.
rl / adapt / train	RL algo flag; LoRA and quantization params; denoiser training hyper-parameters.
data	raw_dir, interim_dir, index_dir, models_dir, reports_dir, trace_path.

6. Running it

```
python scripts/make_demo_corpus.py      # synthetic scanned pages + .txt ground truth in data/raw
python scripts/run_index.py             # Stages 1-4, builds data/index
python scripts/run_eval.py --ablation   # metrics on grading_kit/tasks.jsonl + ablations
uvicorn doc_agent.serve.api:app         # POST /answer?q=...   (or serve.ui.launch(cfg))
python -m pytest                        # 40 tests
python -c "from doc_agent import config; from doc_agent.training import train; train.main('enhancer', config.loa
python -c "from doc_agent import config; from doc_agent.rl import train_rl; train_rl.train(config.load())"
```

Add an LLM key to .env (LLM_API_KEY) to switch from the offline extractive answerer to a real model. Real scans without .txt files make OCR use TrOCR (about 1 GB download on first use).

7. Known limitations

• faiss HNSW segfaults in this environment; flat index is used.

• Layout is classical and sensitive to heavy noise (line retention fell to about 0.18 at moderate noise).

- The enhancer is a one-step denoiser, not a multi-step diffusion model.
- RL trains on a simulator, not on real retrieval traces; decide() itself stays rule-based.
- Grad-CAM, OCR fine-tuning and retriever fine-tuning are not implemented.
- Demo corpus is 10 synthetic pages; the assignment requires 300+ real pages.